

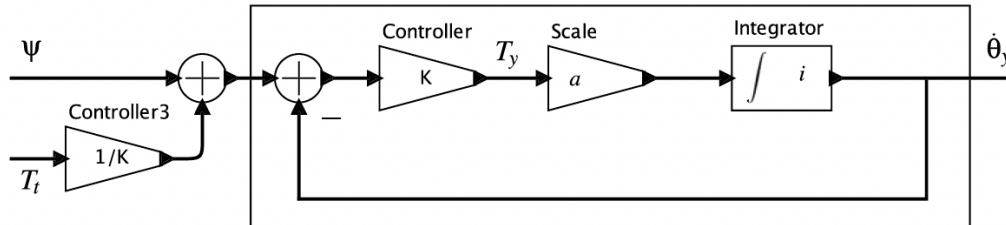
Name: Roll No: e.g. 170001 Dept.: e.g. CSEIndian Institute of Technology Kanpur
CS637 Embedded and Cyber-Physical Systems

Homework Assignment 1

Deadline: September 11, 2026

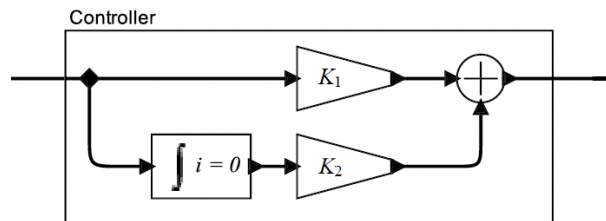
Total: 60 marks**Problem 1.** (10 points)

(a) Construct a Simulink model of the helicopter control system shown in the figure below.



Choose some reasonable parameters and plot the actual angular velocity as a function of time, assuming that the desired angular velocity is zero, $\psi(t) = 0$, and that the top-rotor torque is non-zero, $T_t(t) = bu(t)$. Give your plot for three different values of K and discuss how the behavior varies with K .

(b) Modify the model of part (a) to replace the Controller (the simple scale-by- K actor) with the alternative controller shown in the figure below.



This alternative controller is called a proportional-integrator (PI) controller. It has two parameters K_1 and K_2 . Experiment with the values of these parameters, give some plots of the behavior with the same inputs as in part (a), and discuss the behavior of this controller in contrast to the one of part (a).

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Problem 2. (10 points)

The states of the linearized model of a vehicle steering system represent the lateral deviation of the vehicle from the x -axis and the angle between the vehicle axis and the x -axis. The output of the linearized model is only the first state. Construct a Simulink model for the vehicle steering system with its controller that includes an observer. The dynamics are available in Example 6.4 and Example 7.3 in [AM09]. Apply a sinusoidal signal as the reference trajectory that specifies the desired deviation of the vehicle from the x -axis with time. Plot the output (lateral deviation of the vehicle from the x -axis) with time.

[AM09] K. J. Astrom and R. M. Murray. Feedback Systems: An Introduction for Scientists and Engineers. Princeton University Press, 2009.

http://www.cds.caltech.edu/~murray/books/AM08/pdf/am08-complete_22Feb09.pdf .

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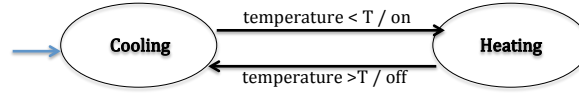
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Problem 3. (10 points)

Consider the following model for a thermostat system.



The thermostat has been designed to maintain the temperature of a room at $T^{\circ}\text{C}$. The model has two states: *cooling* and *heating*. When the system is in the *cooling* state and the temperature of the room goes below $T^{\circ}\text{C}$, the system generates a signal to switch on a heater and moves to the *heating* state. When the temperature of the room goes over $T^{\circ}\text{C}$, the system generates a signal to switch off the heater and moves to the *cooling* state.

- Represent the system as an actor that takes the current temperature as input and produces a signal to control the heater. The actor uses the set point T as a parameter.
- Identify a design problem in the model.
- Provide two different remedies to address the problem.
- Compare your proposed two solutions in terms of ease of implementation and guarantees on the system behavior.

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Problem 4. (10 points)

Construct a timed automation model of a thermostat where the heater will be switched on when the room temperature is less than 22C and switched off when the temperature is above 26C. We also have the following additional requirement. Before switching on or off, it will generate a “beep” sound every second for 5 seconds as a warning. That means once the system senses that the temperature is less than 22C, it will generate the warning first, and then the heater will be switched on. Similarly, a warning sound will be issued before switching off the heater after the temperature reaches 26C.

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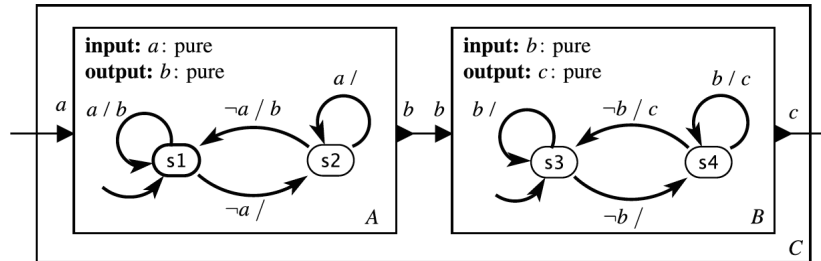
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Problem 5. (10 points)

Consider the following synchronous composition of two state machines A and B.



Construct a single state machine C representing the composition. Which states of the composition are unreachable?

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Problem 6. (10 points)

Consider the following SDF model:



The numbers adjacent to the ports indicate the number of tokens produced or consumed by the actor when it fires. Answer the following questions about this model.

(a) Let q_A , q_B , and q_C denote the number of firings of actors A , B , and C , respectively. Write down the balance equations and find the least positive integer solution.

(b) Find a schedule for an unbounded execution that minimizes the buffer sizes on the two communication channels. What is the resulting size of the buffers?

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